MODULE 02 · DOMAIN 1 - DESIGN APPLICATIONS

From business goal to model task

Write the input and output contract first. Then name the task, then pick the components.

SECTION 1 of 6 EXAM WEIGHT 14% MODULE 02 of 40

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Nobody hires before writing the job description. **Do not pick a model before writing the contract.**

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Five things to write down before anything else

QUESTION	EXAMPLE ANSWER
What arrives, from whom?	A customer's free-text chat message
What must the model see?	Order history, returns policy, product manual
What must come out, for whom?	A reply for the customer, plus a ticket for staff
Under which constraints?	Two seconds, no personal data leaked, low cost
How will we know it works?	Resolved without escalation, measured weekly

ON THE EXAM

Who consumes the output decides its format. A person needs prose. A program needs structure.

From one goal to a set of tasks

01
Goal
Fewer calls to the support line.

02
Request types
Order status, product help, returns.

03
One task each
Look up, answer from docs, fill a form.

04
Components
Tool, retriever, extraction step.

05
Order
Classify first, then hand to the right path.

ON THE EXAM

A business goal is never one task. If a design uses one model call for three kinds of request, one of them is being done badly.

Tasks that understand text

TASK	INPUT TO OUTPUT
Text classification	Text in, one label from a fixed set out
Zero-shot classification	Text plus candidate labels in, best label out, no training
Sentiment analysis	Text in, positive, negative, neutral or mixed out
Token classification, or NER	Text in, named things found inside it out
Sentence similarity	Two texts in, a similarity score out

ON THE EXAM

Understanding tasks return something short and checkable. If the output is a label or a field, you are not generating.

Tasks that produce text

TASK	INPUT TO OUTPUT
Summarisation	Long text in, shorter text with the same gist out
Translation	Text in one language, same meaning in another out
Question answering	Question plus context in, answer grounded in it out
Text2text generation	Text in, transformed text out: rephrase, reformat, correct
Text generation	A prompt in, open-ended continuation out
Speech recognition	Audio in, transcript out

ON THE EXAM

Name the task by what must be preserved. Meaning and length reduced: summarisation. Meaning and language changed: translation.

Summarise or transform? Classify or extract?

SUMMARISATION VS TEXT2TEXT

Is it shorter, with the same gist?

- Shorter, gist kept: summarisation
- Same length, new form: text2text
- Specific when a named task fits
- General bucket only when none does

CLASSIFICATION VS EXTRACTION

Choose, or copy out?

- Pick from a known list: classification
- Pull values from the text: extraction
- Labels are fixed in advance
- Fields are found in the text itself

THE TIEBREAKER

When a specific task name describes the job exactly, it beats the general one. Text2text and text generation are the fallbacks.

Business words, and the task underneath

THE BUSINESS SAYS	THE TASK IS
Route it, tag it, triage it, flag it	Classification
Pull out, capture, populate the fields	Extraction, or NER
Give me the gist, a one-liner, a digest	Summarisation
Answer from our documents	Question answering over retrieval
Find ones like this, spot duplicates	Similarity, using embeddings
Check the system, then do something	Tool calling, perhaps an agent

ON THE EXAM

Scenarios rarely name the task. They describe it in business verbs. Translate the verb before reading the options.

Let the input and output pick the chain

- A question about documents needs a **retriever**, a prompt template and an LLM.
- Several request types need a **classifier or router first**.
- An output read by a program needs a **structured format** and a parser or schema.
- A tone that depends on the user's mood needs a **sentiment step before generation**.
- A live number or an action needs a **tool**. Never ask the LLM to guess it.

> ON THE EXAM
>
> Every component should be traceable to one line of the contract. A component that serves no line is the distractor.

Pennant Outdoor's customer chat, decomposed

01
Message
Free text from a customer.

›

02
Classify intent
Order, product or return?

›

03
Order
Tool: look up the order table.

›

04
Product
Retrieve from manuals, answer.

›

05
Return
Extract fields, open a return.

ON THE EXAM

The router is cheap and fast. It lets each path use the simplest component that does its job.

AI Functions: tasks as SQL functions

FUNCTION	TASK
ai_classify	Classification into labels you supply
ai_extract	Extraction of named fields
ai_summarize	Summarisation
ai_translate	Translation
ai_analyze_sentiment	Sentiment analysis
ai_mask	Masking personal data in text
ai_query	General purpose: your prompt, your chosen model

ON THE EXAM

A task-specific function when the task matches exactly. ai_query when you need your own prompt, model or output schema.

Is anyone waiting for the answer?

SOMEONE IS WAITING

Interactive

- A chat, an app, an API call
- Latency matters per request
- Served from an endpoint or agent
- One input at a time

NOBODY IS WAITING

Batch

- A table of thousands or millions of rows
- Throughput and cost matter most
- AI Functions in a scheduled query
- Results land in a table

THE TIEBREAKER

A million rows and a dashboard tomorrow is batch. One customer and a chat window is interactive. Same task, different components.

> A manufacturer receives warranty claims by email. Staff currently read each one and type the product serial number, the fault description and the purchase date into a claims table.
>
> The team wants a model to do this step instead, so the table fills itself.

Which task category should the engineer evaluate models on?

A Summarisation of each email into a paragraph

B Extraction of named fields from each email

C Sentiment analysis of each customer's tone

D Open-ended generation of a reply to each claim

◆ PAUSE AND DECIDE

Which task category should the engineer evaluate models on?

A Summarisation

Produces a shorter text, not the three separate values the table needs.

B Extraction, token classification

The values are inside the email. Copy them into named fields.

C Sentiment analysis

Nobody asked how the customer feels. That is a different output.

D Text generation

Open-ended output is the opposite of filling fixed columns.

THE SIGNAL

Type the serial number, fault and date into a table. The answer is in the text, and the output is fields. That is extraction.

> A retailer's chatbot uses retrieval over product manuals. It answers product questions well, but fails on messages like where is my order and I want to return this.
>
> About a third of traffic is order status, which lives in an orders table, and a fifth is returns.

What change best addresses the failures?

A. Switch to a model with a larger context window

B. Fine-tune the model on historical order data

C. Export the orders table as text and add it to the manuals index

D. Add an intent classifier that routes each message to the right handler

◆ PAUSE AND DECIDE

What change best addresses the failures?

A **Larger context window**

More room for manual text does not put order data in it.

B **Fine-tune on order data**

Orders change by the minute. Weights would be stale on day one.

C **Orders into the manuals index**

Semantic search is poor at exact lookups, and the text goes stale.

D **Intent classifier and routing**

Three request types, three paths. Orders go to a live lookup.

THE SIGNAL

Works for one request type, fails for the others. That pattern means the design is missing a router, not a better model.

> A support organisation has two million historical tickets in a Unity Catalog table. It wants each ticket tagged with a sentiment and one of eight product categories.
>
> The results feed a dashboard refreshed each morning. The team is mostly SQL analysts.

Which approach fits best?

A Task-specific AI Functions in a scheduled SQL query

B Deploy an agent endpoint and call it once per ticket from a notebook loop

C Build a Knowledge Assistant over the tickets

D Fine-tune a model to predict categories, then serve it

◆ PAUSE AND DECIDE

Which approach fits best?

A AI Functions, scheduled SQL

Batch, table in and table out, SQL skills. Exactly the fit.

B Agent endpoint, loop per ticket

Interactive tooling used for batch: slow, costly, nothing to decide.

C Knowledge Assistant

Answers questions about documents. It does not label rows.

D Fine-tune and serve

Needs labelled data and serving work. Classification needs neither.

THE SIGNAL

Two million rows, a morning dashboard, SQL analysts. Nobody is waiting per ticket: this is batch.

> A mortgage team wants a model to read brokers' emails and create a case in its case-management system through an API.
>
> The business owner's request says: just have it summarise each email. The API expects applicant name, loan amount, property postcode and broker ID.

How should the engineer define the model's output?

A. A short paragraph summarising what each broker's email says

B. The email rewritten in plain English for the case handler

C. JSON with the four API fields, null where a value is absent

D. A priority label for each email, ranked from high down to low

◆ PAUSE AND DECIDE

How should the engineer define the model's output?

A **Paragraph summary**
What was asked for, not what the API can accept.

B **Plain-English rewrite**
Still prose. The API cannot parse a sentence.

C **JSON, four fields, nulls**
Matches the consumer's schema, and says clearly what is missing.

D **Priority label**
A different task. Nothing asked for priority.

THE SIGNAL

The consumer is an API. The output contract comes from the consumer, even when the business describes it in other words.

What to carry forward

- Contract first: **input, context, output and its consumer, constraints, success**.
- Translate the business verb into a task before reading the options.
- Prefer the **specific** task name. Classify chooses; extract copies out.
- Several request types: **route first**. Nobody waiting: **batch**, often AI Functions.
- A program as the consumer means structured output, whatever the request said.

> ON THE EXAM
>
> Next: module 03 writes prompts that reliably return the format the contract demands.